

Answers to 2020 Final Exam (Required Questions)

1. (a) $f'(x) = -4x^{-5} + 6 \sin(2x + 1)$
(b) $f'(x) = 5x^4 \ln(1 + e^{2x}) + x^5 \frac{2e^{2x}}{1 + e^{2x}}$
(c) $f'(x) = (1 + \sin^2 x)^{\sqrt{x}} \left(\frac{1}{2\sqrt{x}} \ln(1 + \sin^2 x) + \sqrt{x} \frac{2 \sin x \cos x}{1 + \sin^2 x} \right)$
2. (a) 1
(b) $e^{-\frac{3}{2}}$
(c) 2
3. $\frac{27!}{6!}$
4. (a) $x(\ln x)^2 - 2(x \ln x - x) + C$
(b) $2(\sqrt{x} + 2) - 4 \ln(\sqrt{x} + 2) + C$
5. (a) $2x \int_x^1 e^{t^2} dt - x^2 e^{x^2}$
(b) $3x^2 \cos(x^8) - \int_0^{x^3} 2t^2 x \sin(t^2 x^2) dt$
6. (a) Diverges
(b) Converges
(c) Converges
7. 62
8. (a) $\int_0^6 \int_0^{3-\frac{y}{2}} f(x, y) dx dy$
(b) $\int_1^e \int_0^{\ln x} f(x, y) dy dx$
9. (a) $\frac{1}{10}$
(b) $\frac{1}{3}(e - 1)$
10. (a) $y(t) = (2e^{\frac{1}{3}t^3} - 1)^{\frac{1}{3}}$
(b) $y(t) = \frac{1}{3}t + \frac{5}{3}t^{-2}$

11. (a) $y(t) = C_1 e^{-4t} + C_2 t e^{-4t}$
(b) $y(t) = C_1 t^{-1} + C_2 t^2$
12. (a) 16
(b) 3